INCH-POUND

MIL-R-39007G
AMENDMENT 2
9 December 1993
SUPERSEDING
AMENDMENT 1
23 March 1993

MILITARY SPECIFICATION

RESISTORS, FIXED, WIRE-WOUND (POWER TYPE),
ESTABLISHED RELIABILITY
GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-R-39007G, dated 16 May 1991, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 2

TABLE II, Designation and Resistance columns, add "R100 to R988 inclusive" and "0.1 to 0.988 inclusive", respectively.

PAGE 6

* 2.1.2, add the following new publication:

"AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B545 (Standard specification for Electrodeposited Coating of Tin).

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)"

PAGE 8

Add new paragraph:

"3.5.1.5.3 Tin plated finishes. Use of tin plating is prohibited as a final finish and as an undercoat effective 6 months from the date of this specification (see 6.9.1). Use of tin-lead (Sn-Pb) finishes are acceptable provided that the minimum lead content is 3 percent."

PAGE 9

TABLE VII, delete and substitute:

"TABLE VII. Resistance temperature characteristic.

| Resistance range | Resistance temperature characteristic parts per million/°C |
|---|---|
| 0.1 R 0.499 | ±650 |
| 0.499 R 1.0 | ±400 |
| 1.0 R 10 | ±50 |
| 10 R | ±20 |

"

PAGE 10

3.22.1, end of first sentence: Delete "±0.5 percent" and substitute "±(0.5 percent +0.005 ohm)."

3.22.2, end of first sentence: Delete "±1.0 percent" and substitute "±(1.0 percent +0.005 ohm)."


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*3.23, delete "±0.5 percent" (two places) and substitute "±1 percent" (two places).

PAGE 12

Add the following new paragraph,

"3.26.4 Beryllium oxide (BeO). Manufacturers which use beryllium oxide in their construction shall mark each resistor body and resistor package with the symbol "BeO"."

PAGE 16

4.4.4.2, add the following sentence: "In the event that the contractor meets or exceeds 100 ppm for ppm-2 the qualifying activity shall take the actions specified in EIA-554, method B."

PAGE 18

*TABLE XII, Subgroup 2: Delete "4.7.4" and substitute "4.7.3".
Subgroup 3: Delete "Dielectric withstanding voltage" and all associated information.

*4.6.1.2.1, first sentence, delete and substitute:

"Subgroup 1. Subgroup 1 tests shall be performed on a production lot basis on 100 percent of the product supplied under this specification."

PAGE 20

4.6.1.2.5.2, last sentence, delete and substitute:

"If the lot fails this solderability test the lot may be reworked a second time and retested. If the lot fails the second rework, the lot shall be considered rejected and shall not be furnished against the requirements of the specification."

PAGE 21

*TABLE XIV, delete and substitute:

"TABLE XIV. Group B inspection.

| Inspection | Requirement paragraph | Test method paragraph | Number of sample |
|---|---|---|---|
| Resistance to solvents 1/ | 3.11 | 4.7.5 | 12 |

1/ If the manufacturer can demonstrate that this test has been performed for five consecutive times with zero failures, the frequency of this test, with the approval of the qualifying activity, can be performed on an annual basis. If the design, material, construction or processing of the part is changed, or if there are any quality problems or failures, the qualifying activity may require resumption of the original test frequency."

*After 4.6.2.1.1.2, add new paragraph:

"4.6.2.1.1.2.1 Quarterly. Every 3 months, 10 sample units of each style and characteristic and of any resistance value shall be subjected to the tests specified in table XV, in the order shown."



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*TABLE XV, delete and substitute:

* TABLE XV. Group C inspection.

| Inspection | Requirement paragraph | Test method paragraph | Number of sample | Number of defects allowed |
|---|---|---|---|---|
| Monthly | | | | |
| Subgroup 1 | | | | |
| Life | 3.22 | 4.7.16 | See 4.6.2.1.1.1 | See 4.6.2.1.1.1 |
| Subgroup 2 | | | | |
| Resistance temperature characteristic 1/ | 3.13 | 4.7.7 | 10 (6 highest 4 lowest) | 1 |
| Moisture resistance | 3.18 | 4.7.12 | | |
| Quarterly 2/ | | | | |
| Dielectric withstanding voltage | 3.16 | 4.7.10 | | |
| Insulation resistance | 3.17 | 4.7.11 | 10 | |
| Low temperature storage | 3.14 | 4.7.8 | (5 highest | 1 |
| Terminal strength | 3.19 | 4.7.13 | 5 lowest) | |
| Semiannually 2/ | | | | |
| Subgroup 1 | | | | |
| Short time overload | 3.15 | 4.7.9 | | |
| Thermal shock | 3.12 | 4.7.6 | 30 | 1 |
| Shock (specified pulse) | 3.20 | 4.7.14 | (15 highest | |
| Vibration (high frequency) | 3.21 | 4.7.15 | 15 lowest) | |
| Subgroup 2 2/ | | | (10 highest 30 10 3/ | |
| Reactance | 3.25 | 4.7.19 | 10 lowest) | 0 |
| Annual | | | 102 (51 highest | |
| High temperature exposure | 3.23 | 4.7.17 | 51 lowest) | 1 |

1/ If the manufacturer can demonstrate that this test has been performed for five consecutive times with zero failures, the frequency of this test, with the approval of the qualifying activity, can be performed on an annual basis. If the design, material, construction or processing of the part is changed, or if there are any quality problems or failures, the qualifying activity may require resumption of the original test frequency.

2/ If the manufacturer can demonstrate that these tests have been performed for five consecutive times with zero failures, these tests, with the approval of the quality activity, can be deleted. The manufacturer, however, shall perform these tests every 3 years after the deletion as part of long term design verification. If the design, material, construction or processing of the part is changed, or if there are any problems, the qualifying activity may require resumption of the specified testing. Deletion of testing does not relieve the manufacturer from meeting the test requirement in case of dispute.

3/ Ten samples of values representative of production in that period.*



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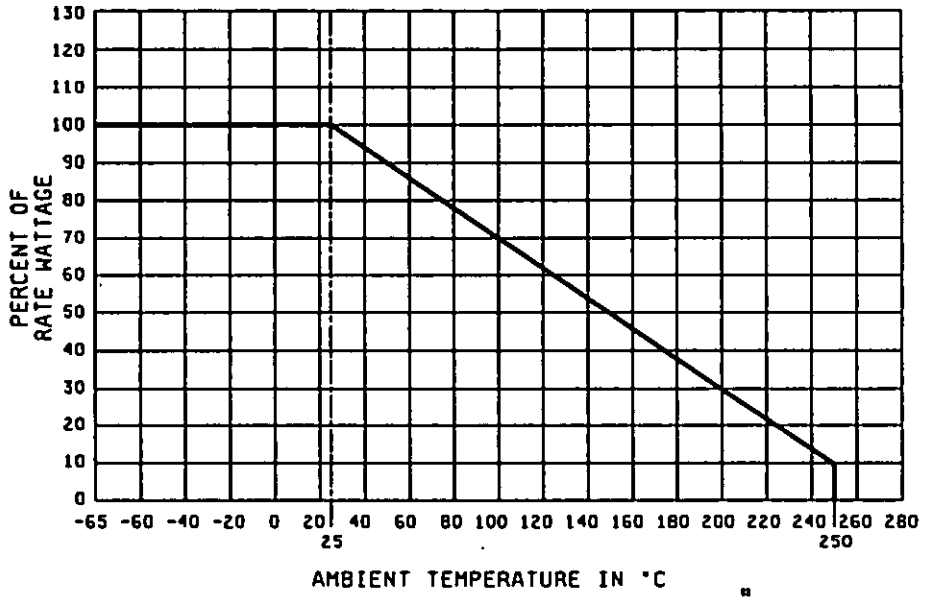
PAGE 29

*4.7.17c.: Delete "+275°C" and substitute "+250°C".

PAGE 37

*FIGURE 6, delete and substitute:

"




"

PAGE 38

6.7, add to end of last sentence, "and TABLE XVI-1 within".

Add new paragraph:

"6.9.1 Tin plated finishes. Tin plating is prohibited (see 3.5.1.5.3) since it may result in tin whisker growth. Tin whisker growth could adversely affect the operation of electronic equipment systems. For additional information on this matter refer to ASTM B545 (Standard Specification for Electrodeposited Coating of Tin)."

Add new TABLE XVI-1:



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"TABLE XVI-1. Performance requirements.

| Style | | RWR71 1/ | RWR74 1/ | RWR78 | RWR80 | RWR81 | RWR84 | RWR89 | RWR82 |
|---|---|---|---|---|---|---|---|---|---|
| Maximum resistance temperature characteristic in parts/million/°C-ppm (Reference to 25°C) | 0.1 ≤ r ≤ 0.499 | +650 ppm | +650 ppm | +650 ppm | +650 ppm | +650 ppm | +650 ppm | +650 ppm | +650 ppm |
| | 0.499 < R ≤ 1.0 | +400 ppm | +400 ppm | +400 ppm | +400 ppm | +400 ppm | +400 ppm | +400 ppm | +400 ppm |
| | 1.0 < R 10 | ±50 ppm | ±50 ppm | ±50 ppm | ±50 ppm | ±50 ppm | ±50 ppm | ±50 ppm | ±50 ppm |
| | 10 < R | ±20 ppm | ±20 ppm | ±20 ppm | ±20 ppm | ±20 ppm | ±20 ppm | ±20 ppm | ±20 ppm |
| Maximum ambient temperature at rated wattage (see figure 6) | | 25°C | 25°C | 25°C | 25°C | 25°C | 25°C | 25°C | 25°C |
| Maximum ambient temperature at zero wattage derating (see figure 6) | | 275°C | 275°C | 275°C | 275°C | 275°C | 275°C | 275°C | 275°C |
| Power rating in watts | | 2 | 5 | 10 | 2 | 1 | 7 | 3 | 1.5 |
| Maximum percent change in resistance ± 2/ | Conditioning (see 3.8) | .2 | .2 | .2 | .2 | .2 | .2 | .2 | .2 |
| | Thermal shock (see 3.12) | .2 | .2 | .2 | .2 | .2 | .2 | .2 | .2 |
| | Low temperature storage (see 3.14) | .1 | .1 | .1 | .1 | .1 | .1 | .1 | .1 |
| | Short-time overload (see 3.15) | .2 | .2 | .2 | .2 | .2 | .2 | .2 | .2 |
| | Dielectric withstanding voltage (see 3.16) | .1 | .1 | .1 | .1 | .1 | .1 | .1 | .1 |
| | Moisture resistance (see 3.18) | .2 | .2 | .2 | .2 | .2 | .2 | .2 | .2 |
| | Shock (specified pulse) (see 3.20) | .1 | .1 | .1 | .1 | .1 | .1 | .1 | .1 |
| | Vibration, high frequency )see 3.21) | .1 | .1 | .1 | .1 | .1 | .1 | .1 | .1 |
| | Life (see 3.22) | see 3.22 | see 3.22 | see 3.22 | see 3.22 | see 3.22 | see 3.22 | see 3.22 | see 3.22 |
| | High temperature exposure (see 3.23) | see 3.23 | see 3.23 | see 3.23 | see 3.23 | see 3.23 | see 3.23 | see 3.23 | see 3.23 |
| | Insulation resistance (see 3.17) | 1,000 megohms min | 1,000 megohms min | 1,000 megohms min | 1,000 megohms min | 1,000 megohms min | 1,000 megohms min | 1,000 megohms min | 1,000 megohms min |
| | Insulation resistance (after moisture resistance) (see 3.18) | 100 megohms min | 100 megohms min | 100 megohms min | 100 megohms min | 100 megohms min | 100 megohms min | 100 megohms min | 100 megohms min |
| Resistance tolerances ± percent (see table IV) | | 1,.5,.1 | 1,.5,.1 | 1,.5,.1 | 1,.5,.1 | 1,.5,.1 | 1,.5,.1 | 1,.5,.1 | 1,.5,.1 |

1/ Not for use on new design; for replacement purposes only.
2/ Where total resistance change is 1 percent or less, it shall be considered as ±(___ percent +0.005 ohm)."





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## CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Navy - AS, CG, MC, OS
Air Force - 17, 19, 80, 85
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

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